

Monitoring of Processes and Threads

Exercises 1

a)

```
~/code/university/semester4/rt/linux_exercises/monitoring_oPat >> ps -u $(whoami) -o ppid,pid,psr,sgl_p,pcpu,comm,policy,rtprio,prti,nice,time,c,f,w
PPID  PID  PSR  P  %CPU COMMAND      POL  RTPRIO  PRI  NI   TIME  C  F  WCHAN  CMD      %MEM  MAJFL  MINFL  SZ
1      3314  6 *  0.2 systemd      TS   -  19  0 00:00:01 0 4 -   /usr/lib/systemd/systemd -- 0.0    0    5000  5959
3314   3316 12 *  0.0 (sd-pam)      TS   -  19  0 00:00:00 0 5 -   (sd-pam) 0.0    0     58  5799
3314   3424 12 *  0.1 dbus-daemon  TS   -  19  0 00:00:00 0 0 ep_pol /usr/bin/dbus-daemon --sess 0.0    0    2567  2490
3314   3425 1 *  0.1 pipewire  TS   -  30 -11 00:00:00 0 0 ep_pol /usr/bin/pipewire 0.0    5   17765 31664
3314   3428 6 *  0.1 user-session-he TS -  19  0 00:00:00 0 4 do_wai /snap/snapd-desktop-integra 0.0    56   8434 75901
3314   3429 5 *  0.0 gnome-keyring-d TS -  19  0 00:00:00 0 0 poll_s /usr/bin/gnome-keyring-daem 0.0    38   1084 48630
3314   3435 0 *  0.0 mpris-proxy   TS   -  19  0 00:00:00 0 0 poll_s /usr/bin/mpiris-proxy 0.0    1     787  1923
3314   3441 15 * 0.2 wireplumber TS -  30 -11 00:00:01 0 0 poll_s /usr/bin/wireplumber 0.1    1   4304 184399
3314   3442 0 *  0.0 pipewire     TS   -  19  0 00:00:00 0 0 ep_pol /usr/bin/pipewire -c filter 0.0    0   1013 24024
3314   3443 5 *  0.0 pipewire-pulse TS -  30 -11 00:00:00 0 0 ep_pol /usr/bin/pipewire-pulse 0.0    2   16493 52026
1      3486 14 * 0.0 ksecret      TS   -  19  0 00:00:00 0 0 poll_s /usr/bin/ksecret --pam-log 0.5    23   6685 389056
3294   3487 6 *  0.0 startplasma-way TS -  19  0 00:00:00 0 4 poll_s /usr/bin/startplasma-waylan 0.1    16   3813 63720
3314   3574 3 *  0.0 xdg-document-po TS -  19  0 00:00:00 0 0 poll_s /usr/libexec/xdg-document-p 0.0    0     950 137728
3314   3589 4 *  0.0 xdg-permission- TS -  19  0 00:00:00 0 0 poll_s /usr/libexec/xdg-permission 0.0    0     883 79662
3314   3714 4 *  0.0 gcr-ssh-agent TS -  19  0 00:00:00 0 0 poll_s /usr/libexec/gcr-ssh-agent 0.0    10     973 24388
3314   3717 2 *  0.0 kwln_wayland_wr TS -  19  0 00:00:00 0 0 poll_s /usr/bin/kwln_wayland_wrapp 0.0    1   1395 48552
3314   3719 8 *  0.0 ssh-agent    TS   -  19  0 00:00:00 0 0 -   /usr/bin/ssh-agent -D 0.0    1     980 2621
3717   3725 14 * 4.4 kwln_wayland BB 1 41 - 00:00:23 4 4 -   /usr/bin/kwln_wayland --way 1.0    65 183493 858563

~/code/university/semester4/rt/linux_exercises/monitoring_oPat >> ps -u $(whoami) -o ppid,pid,psr,sgl_p,pcpu,comm,policy,rtprio,prti,nice,time,c,f,w
3314   7657 14 14 99.9 load_1 TS - 0 19 00:03:13 99 0 - ./load_1 1 20 19-1 500000000 0.0 0 120 693
3314   7658 11 11 100 load_2 TS - 0 19 00:03:13 99 0 - ./load_2 2 20 19-2 500000000 0.0 0 119 693
3314   7659 8 8 100 load_3 TS - 0 19 00:03:13 99 0 - ./load_3 3 20 19-3 500000000 0.0 0 119 693
3314   7660 3 3 99.9 load_4 TS - 0 19 00:03:13 99 0 - ./load_4 4 20 19-4 500000000 0.0 0 117 693
3314   7661 7 7 99.9 load_5 TS - 0 19 00:03:13 99 0 - ./load_5 5 20 19-5 500000000 0.0 0 119 693
5960   8189 15 * 0.0 grep TS - 19 0 00:00:00 0 0 anon_p grep --color=auto -l load 0.0 0 162 4501
```

Flag	Meaning
-u	user
-o	Display the following columns
PPID	parent process id
PID	process id
PSR	designatedCPU core number
SGI_P	cpu core its actually running on rn
%CPU	cpu core usage in %
COMM	full command without args
POLICY	scheduling policy
RTPRIO	Real time priority
PRI	Prio of the process
NICE	niceness
TIME	used cpu time
C	full cpu usage value
F	process flags (for kernel)
WCHAN	what sleep function from the kernel the process is using
CMD	full command with args
%MEM	memory usage in %
MAJFLT	major page faults
MINFLT	minor page faults
SZ	size of process given in pages (4kb)

b)

```
~/code/uni/semester4/rts/linux_exercises/monitoring_oPaT >> pstree -acghlpsUu
systemd,1,1 --switched-root --system --deserialize=52 splash
├─ModemManager,1861,1861
│   └─{ModemManager},1869,1861
│       └─{ModemManager},1870,1861
│           └─{ModemManager},1873,1861
├─NetworkManager,1755,1755 --no-daemon
│   └─{NetworkManager},1855,1755
│       └─{NetworkManager},1857,1755
│           └─{NetworkManager},1858,1755
├─accounts-daemon,1657,1657
│   └─{accounts-daemon},1718,1657
│       └─{accounts-daemon},1719,1657
│           └─{accounts-daemon},1749,1657
├─avahi-daemon,1619,1619,avahi
│   └─avahi-daemon,1682,1619
├─bluetoothd,1620,1620
├─chronyd-starter,1622,1622 /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
│   └─chronyd,1743,1622, _chrony -n -F 1
│       └─chronyd,1754,1622 -n -F 1
├─colord,3278,3278,colord
│   └─{colord},3282,3278
│       └─{colord},3283,3278
│           └─{colord},3285,3278
├─containerd,1971,1971
│   └─{containerd},2021,1971
│       └─{containerd},2022,1971
├─load_1,7657,7647 1 20 19-1 5000000000
├─load_2,7658,7647 2 20 19-2 5000000000
├─load_3,7659,7647 3 20 19-3 5000000000
├─load_4,7660,7647 4 20 19-4 5000000000
├─load_5,7661,7647 5 20 19-5 5000000000
└─grep,8503,8502 --color=auto -i load
```

Flag	Meaning
-a	Shows command line args
-c	doesn't truncate branches, you see everything
-g	shows group ids
-h	highlights parent process
-l	doesn't truncate long strings
-p	shows pids
-s	shows process inheritance
-U	uses unicode
-u	shows different users

c)

The command `ps -ef` shows all processes detailed

```

4371f050-1cc9a0bc math ~$ math
~/code/uni/semester4/rts/linux_exercises/monitoring_oPaT ~$ ps -ef | grep sgma
root      3294      2292  0 14:13 ?        00:00:00 /usr/lib/x86_64-linux-gnu/sddm/sddm-helper
/usr/bin/startplasma-wayland --user sgma
sgma      3314      1  0 14:13 ?        00:00:01 /usr/lib/systemd/systemd --user
sgma      3316      3314  0 14:13 ?        00:00:00 (sd-pam)
sgma      3424      3314  0 14:13 ?        00:00:00 /usr/bin/dbus-daemon --session --address=
sgma      3425      3314  0 14:13 ?        00:00:00 /usr/bin/pipewire
sgma      3428      3314  0 14:13 ?        00:00:00 /snap/snapd-desktop-integration/387/usr/
sgma      3429      3314  0 14:13 ?        00:00:00 /usr/bin/gnome-keyring-daemon --foregrou
sgma      3435      3314  0 14:13 ?        00:00:00 /usr/bin/mpris-proxy
sgma      3441      3314  0 14:13 ?        00:00:02 /usr/bin/wireplumber
sgma      3442      3314  0 14:13 ?        00:00:00 /usr/bin/pipewire -c filter-chain.conf
sgma      3443      3314  0 14:13 ?        00:00:00 /usr/bin/pipewire-pulse
sgma      3486      1  0 14:13 ?        00:00:00 /usr/bin/ksecretd --pam-login 13 14
sgma      3487      3294  0 14:13 tty2    00:00:00 /usr/bin/startplasma-wayland
sgma      3574      3314  0 14:13 ?        00:00:00 /usr/libexec/xdg-document-portal
sgma      3589      3314  0 14:13 ?        00:00:00 /usr/libexec/xdg-permission-store
sgma      3714      3314  0 14:13 ?        00:00:00 /usr/libexec/gcr-ssh-agent --base-dir /r
sgma      3717      3314  0 14:13 ?        00:00:00 /usr/bin/kwin_wayland_wrapper --xwayland
sgma      3719      3314  0 14:13 ?        00:00:00 /usr/bin/ssh-agent -D
sgma      3735      3717  3 14:13 ?        00:00:36 /usr/bin/kwin_wayland --wayland-fd 7 --s
sgma      3736      3314  0 14:13 ?        00:00:00 /usr/libexec/gvfsd
sgma      3756      3314  0 14:13 ?        00:00:00 /usr/libexec/gvfsd-fuse /run/user/1000/gv
sgma      3834      3735  0 14:13 ?        00:00:00 /usr/bin/Xwayland :0 -auth /run/user/1000
sgma      3884      3314  0 14:13 ?        00:00:00 /usr/libexec/at-spi-bus-launcher
sgma      3918      3884  0 14:13 ?        00:00:00 /usr/bin/dbus-daemon --config-file=/usr/

2cf3c091-732695c16 math ~$ math
~/code/uni/semester4/rts/linux_exercises/monitoring_oPaT ~$ ps -ef | grep -i "load"
root      1609      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1610      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1611      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1612      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1613      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1614      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1615      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
root      1616      2  0 14:13 ?        00:00:00 [kworker/R-amdgpu_dm_hpd_rx_offload_v
sgma      7657      3314  99 14:19 pts/2    00:18:35 ./load_1 1 20 19-1 5000000000
sgma      7658      3314  99 14:19 pts/2    00:18:36 ./load_2 2 20 19-2 5000000000
sgma      7659      3314  99 14:19 pts/2    00:18:35 ./load_3 3 20 19-3 5000000000
sgma      7660      3314  99 14:19 pts/2    00:18:35 ./load_4 4 20 19-4 5000000000
sgma      7661      3314  99 14:19 pts/2    00:18:35 ./load_5 5 20 19-5 5000000000
sgma      9018      8904  0 14:30 ?        00:00:00 bwrap --unshare-all --die-with-parent
fs /tmp/.run --clearenv --setenv HOME /tmp/home --setenv XDG_RUNTIME_DIR /tmp/.run --setenv

```

Minor page fault

A minor page fault happens when the program tries to access a page that is not currently loaded into its virtual address space. When the page is outside this address space but still inside the RAM, a minor page fault occurs, which means that the system needs to update the page table.

This only costs little time and effort compared to a major page fault, where the requested page is not only outside the virtual address space, but also outside the RAM (on a harddisk).

This means that the system needs to swap from the harddrive, which is very time and resource expensive.